

P2 Cholesky factorisation

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In this note, we wish to prove any positive semi-definite Hermitian matrix H can be written as a product of a lower diagonal matrix $A^\dagger$ and its' Hermitian adjoint A , which is an upper diagonal matrix (used in one of the subroutines for fitting eco frequencies)

$$H = A^\dagger A \quad (1)$$

which seems familiar to factorising the quantum harmonic oscillator Hamiltonian (minus a zero point energy) into its ladder operators, where A is the annihilation operator. H is Hermitian, so it has the following spectral expansion

$$H = \sum_i |i\rangle E_i \langle i| \quad (2)$$

trivially, there exist $B = \sum_i |i\rangle \sqrt{E_i} \langle i|$, ($\sqrt{E_i}$ is real since $E_i = \langle i|H|i\rangle \geq 0$) such that $B^\dagger B = H$, in fact B is also Hermitian. B is diagonal in the eigenbasis of H , but needs not to be diagonal in an arbitrary basis $\{|e_i\rangle\}$. We want to find the unitary transformation of B so it becomes an upper diagonal matrix. Define

$$|b_j\rangle = B|e_j\rangle \quad (3)$$

then construct an orthonormal basis $\{|u_i\rangle\}$ by removing the components of $|b_i\rangle$ that sticks into the subspace spanned by the previous vectors, for the first vector, we just need to normalise $|u_1\rangle = |b_1\rangle / \sqrt{\langle b_1|b_1\rangle}$ then by induction

$$|u'_j\rangle = |b_j\rangle - \sum_{i < j} |u_i\rangle \langle u_i|b_j\rangle \quad (4)$$

normalise $|u_j\rangle = |u'_j\rangle / \sqrt{\langle u'_j|u'_j\rangle}$, if $|u'_j\rangle = 0$, choose a unit vector orthogonal to all previous vector, then we can construct a unitary operator

$$U = \sum_i |u_i\rangle \langle e_i| \quad (5)$$

then define

$$A = U^\dagger B \quad (6)$$

the matrix element in the original basis is

$$A_{ij} = \langle e_i|A|e_j\rangle = \sum_k \langle e_i|e_k\rangle \langle u_k|B|e_j\rangle = \sum_k \langle e_i|e_k\rangle \langle u_k|b_j\rangle = \langle u_i|b_j\rangle \quad (7)$$

but $\langle u_i|b_j\rangle = 0$ for $i > j$, thus $A_{ij} = 0$ for $i > j$, which is same as saying A is upper triangular, and finally

$$A^\dagger A = B^\dagger U U^\dagger B = B^\dagger B = H \quad (8)$$